

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Phase 1, Open-label, Non-comparative, Multicenter Clinical Study to Evaluate the Safety, Tolerability, and Pharmacokinetics of Ceftolozane/Tazobactam (MK-7625A) in Pediatric Participants With Nosocomial Pneumonia **Short Title/Acronym:** Safety and PK of Ceftolozane/Tazobactam in Pediatric Nosocomial Pneumonia (MK-7625A-036) **Protocol Number:** MK-7625A-036 **NCT Number:** NCT04223752 **Sponsor Name:** Merck **Company:** Merck **Investigational Product:** Ceftolozane/Tazobactam (MK-7625A) **Drug Preferred Name:** Tazobactam **Trade Names:** Zosyn, Zerbaxa **ATC Code:** J01CG02 **Phase of Development:** Phase I **Trial Status:** COMPLETED **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/zerbaxa-epar-product-information_en.pdf **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety and tolerability of ceftolozane/tazobactam (MK-7625A) by measuring the percentage of pediatric participants experiencing any Adverse Events (AEs), Serious Adverse Events (SAEs), or Drug-related AEs. **Secondary Objectives:** To evaluate the pharmacokinetics of plasma ceftolozane, specifically tracking Plasma Concentrations, Area Under the Concentration-time Curve of an 8-hour Dosing Interval (AUC₀₋₈), and Maximum Observed Concentration (C_{max}).

2.2 Background & Rationale To establish the safety, tolerability, and pharmacokinetic (PK) profiles of ceftolozane/tazobactam in pediatric populations. Children and infants hospitalized with nosocomial pneumonia (NP) represent a vulnerable population requiring precise weight- and age-adjusted dosing regimens to ensure effective bacterial eradication and minimize toxicity.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Phase**

Requirement: Trial must be in PHASE1 **Control Type:** Non-comparative (Single-arm) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** 41 evaluable pediatric participants **Number of Sites:** Multicenter **Estimated Study Start:** N/A (Study is Completed) **Estimated Primary Completion:** N/A (Study is Completed)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Is hospitalized and anticipated to receive a minimum of 8 days of concomitant standard-of-care (SOC) antibiotic therapy for proven or suspected Nosocomial Pneumonia (NP). 2. If male, is abstinent from heterosexual intercourse, or agrees to use contraception during the intervention period and for ≥ 30 days after the last dose of study intervention. 3. If female, is not pregnant or breastfeeding, or is not a woman of childbearing potential (WOCBP), or is a WOCBP using acceptable contraception, has a negative urine or serum pregnancy test within 48 hours of the first dose of study intervention, or is abstinent from heterosexual intercourse.

4.2 Exclusion Criteria 1. Has a documented history of any moderate or severe hypersensitivity (or allergic) reaction to any β -lactam antibacterial. 2. Participants 3 months to <18 years of age with moderate to severe impairment of renal function (CrCL <50 mL/min/1.73 m²), or participants <3 months of age with CrCL <20 mL/min/1.73 m², or any requirement for peritoneal dialysis, hemodialysis, or hemofiltration. 3. Is receiving or is anticipated to receive piperacillin/tazobactam while receiving ceftolozane/tazobactam, or has received it within 24 hours prior to the first dose. 4. Has participated in any clinical study of a therapeutic investigational product within 30 days prior to the first dose. 5. Has previous participation in any study of ceftolozane or ceftolozane/tazobactam. 6. Has any condition or circumstance that, in the opinion of the investigator, would compromise safety or data quality, including rapidly progressing disease or immediately life-threatening illness (e.g., acute hepatic failure, septic shock), or active immunosuppression.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ceftolozane/tazobactam (MK-7625A) administered at pediatric-adjusted dosages across an 8-hour dosing interval. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Concomitant with standard-of-care therapy (minimum of 8 days of SOC antibiotic therapy).

5.2 Concomitant & Prohibited Therapy Permitted: Concomitant standard-of-care (SOC) antibiotic therapy for proven or suspected NP. **Prohibited:** Piperacillin/tazobactam usage during the study or within 24 hours prior to the first dose; active immunosuppression therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -1 to 0 Informed Consent,

Medical History, Pregnancy Test, Renal Function (CrCL) Assessment | | **Baseline** | Day 0 | First Dose, PK Sampling Initiation | | **Interim** | Days 1 to 8+ | Inpatient Safety Monitoring, PK Blood Draws (AUC₀₋₈, C_{max}), SOC administration | | **End of Study** | Day 30 post-dose | Final Safety Evaluation, AE/SAE Resolution Check, Contraception Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Percentage of Participants With Any Adverse Events (AEs):** An AE is defined as any untoward medical occurrence temporally associated with the use of study intervention, whether or not considered related.
- 2. Percentage of Participants With Any Serious AEs (SAEs):** Defined as any untoward medical consequence resulting in death, life-threatening scenarios, inpatient hospitalization or prolongation thereof, persistent/significant disability, or congenital anomaly.
- 3. Percentage of Participants With Any Drug-related AEs:** Defined as AEs temporally associated with the study intervention and considered related to it.

7.2 Secondary Endpoints

1. Plasma Concentrations of Ceftolozane.
2. Area Under the Concentration-time Curve of an 8-hour Dosing Interval (AUC₀₋₈) of Plasma Ceftolozane.
3. Maximum Observed Concentration During a Dosage Interval (C_{max}) of Plasma Ceftolozane.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard inpatient monitoring capturing any untoward medical occurrence in a clinical study participant. **Serious Adverse Events (SAEs):** Reported immediately for events resulting in death, life-threatening conditions, prolongation of hospitalization, or significant disability/incapacity. **Data Monitoring Committee (DMC):** Internal safety review committee governed by the Sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving at least one dose of study medication for AE reporting) and PK Evaluable Population. **Sample Size Justification:** N=41\$, determined adequate for Phase 1 open-label pharmacokinetic modeling and initial safety characterization in pediatric cohorts. **Handling of Missing Data:** Standard pharmacokinetic imputation rules for missing plasma concentration timepoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Continuous inpatient monitoring during the dosing phase; standard Phase 1 clinical site auditing for PK sampling accuracy. **Audit Trail:** Electronic Case Report Form (eCRF) locking and stringent data clarification protocols.

11. Study Identifiers & Governance

Posting ID: 684063392650693124 **Registry Number:** NCT04223752

Sponsor/Lead Organization: Merck **Study Title:** Safety and Pharmacokinetics of Ceftolozane/Tazobactam in Pediatric Participants With Nosocomial Pneumonia (MK-7625A-036) **Official Regulatory Title:** A Phase 1, Open-label, Non-comparative, Multicenter Clinical Study to Evaluate the Safety, Tolerability, and Pharmacokinetics of Ceftolozane/Tazobactam (MK-7625A) in Pediatric Participants With Nosocomial Pneumonia

12. Clinical Framework (Nosocomial Pneumonia Specific)

Disease Areas: Nosocomial Pneumonia **Primary Condition:** Nosocomial Pneumonia (Preferred Name: Hospital acquired pneumonia)

Definition: Pneumonia acquired during a hospital stay. **Semantic**

Type: Disease or Syndrome **MeSH Code:** D000077299 **SNOMED ID:**

425464007 **Diagnosis Threshold:** Proven or suspected NP requiring a minimum of 8 days of standard-of-care antibiotics **Medical**

Methodology: Investigational PK profiling alongside Standard-of-Care (SOC) antibiotic therapy **Optimization Target:**

Pharmacokinetic parameters (Cmax and AUC0-8) and safety tolerability

13. Study Procedures & Methodology

Management Pathway: Standard inpatient clinical pathway for pediatric NP **Education Protocol (HCP):** Phase 1 PK sampling instruction and trial-specific safety reporting **Education**

Protocol (Patient): N/A (Pediatric/Inpatient study; consent via parent/guardian) **Quality Audit Mechanism:** Real-time inpatient PK

collection audits and strict AE/SAE categorization tracking

14. Observation & Follow-Up Schedule

Total Duration: Intervention period + ≥ 30 days safety

follow-up **Onsite Visit Cadence:** Daily while hospitalized **Remote**

Monitoring Frequency: N/A (Inpatient setting) **Checklist Review**

Interval: Continuous clinical observation during dosing interval

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					